PROJECT PRESENTATION

ADVENTURE WORKS

Sales Analysis & Dashboard Project

End-to-End Data Analytics using
Excel • SQL • Power BI • Tableau

Data Preparation • Analysis • DAX • Dashboards • Business Insights

MICROSOFT EXCEL

POWER BI

SQL

TABLEAU

GROUP 2

PROJECT OVERVIEW

Adventure Works Sales Analytics • Excel • SQL • Power BI • Tableau

Adventure Works Cycles is a fictional multinational bicycle manufacturer and seller. This project transforms raw sales data into business insights across products, customers, regions and time, using Excel, SQL, Power BI and Tableau.

DATA PREPARATION

Append and merge sales/product/customer data; clean fields and create relationships.

SQL ANALYSIS

13 business questions solved with queries, joins, dates and aggregations.

VISUALIZATION

Excel, Tableau and Power BI dashboards with KPIs, charts and slicers.

POWER QUERY

Data cleaning, append/merge, transformations and calculated fields.

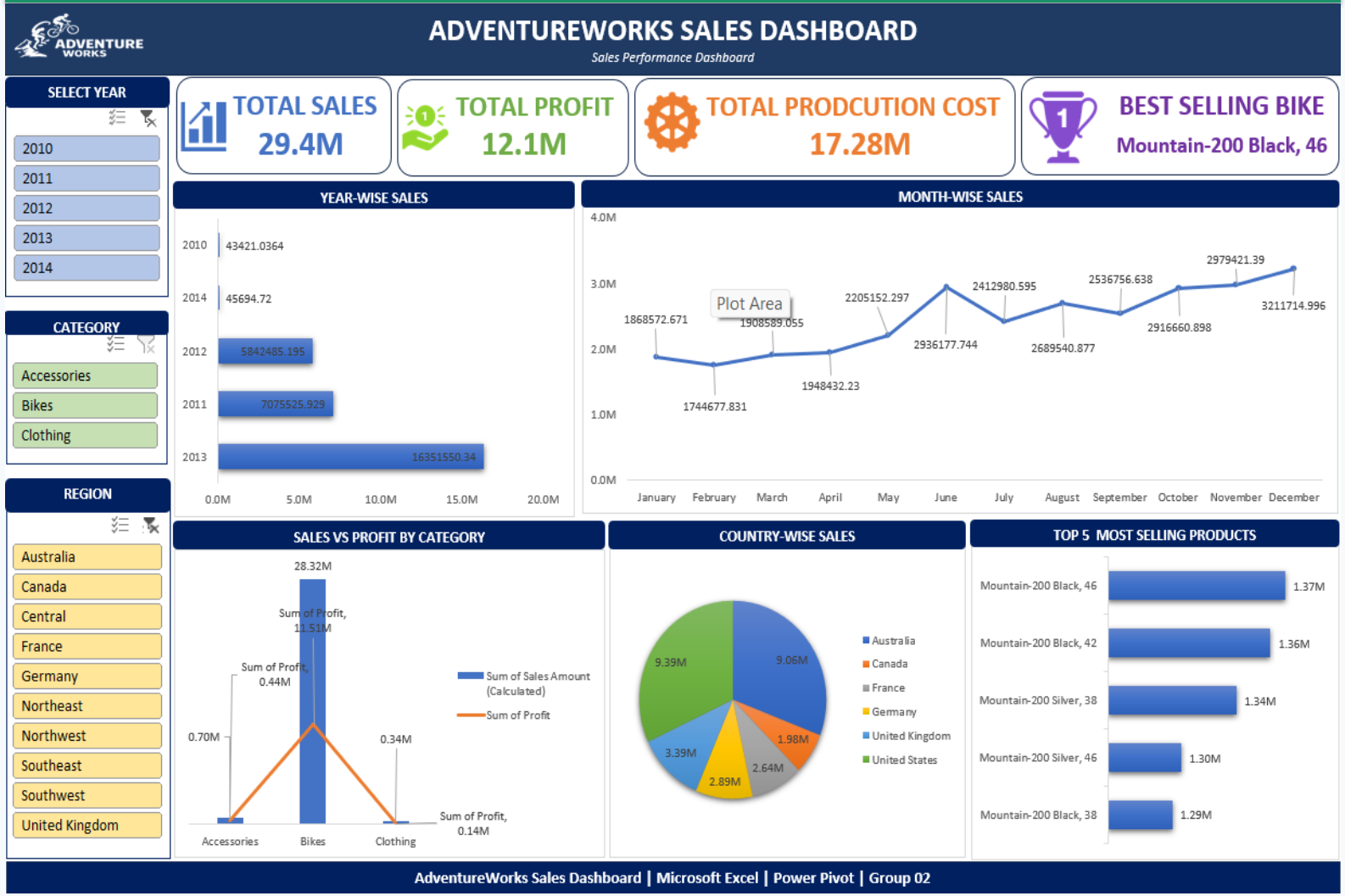
PROJECT WORKFLOW

Raw Data → Data Prep → Model → Calculations → Dashboard → Insights

DATA MODEL : Fact_Sales → DimCustomer | DimProduct | DimDate | DimSalesTerritory

EXCEL DASHBOARD

EXCEL DASHBOARD SCREENSHOT



Pivot Tables

Year-wise, month-wise, quarter-wise and region analysis.

KPI Cards

Total Sales, Total Profit, Total Orders, Avg Order and Profit Margin.

Interactive Slicers

Year, Region and Category filters for focused analysis.

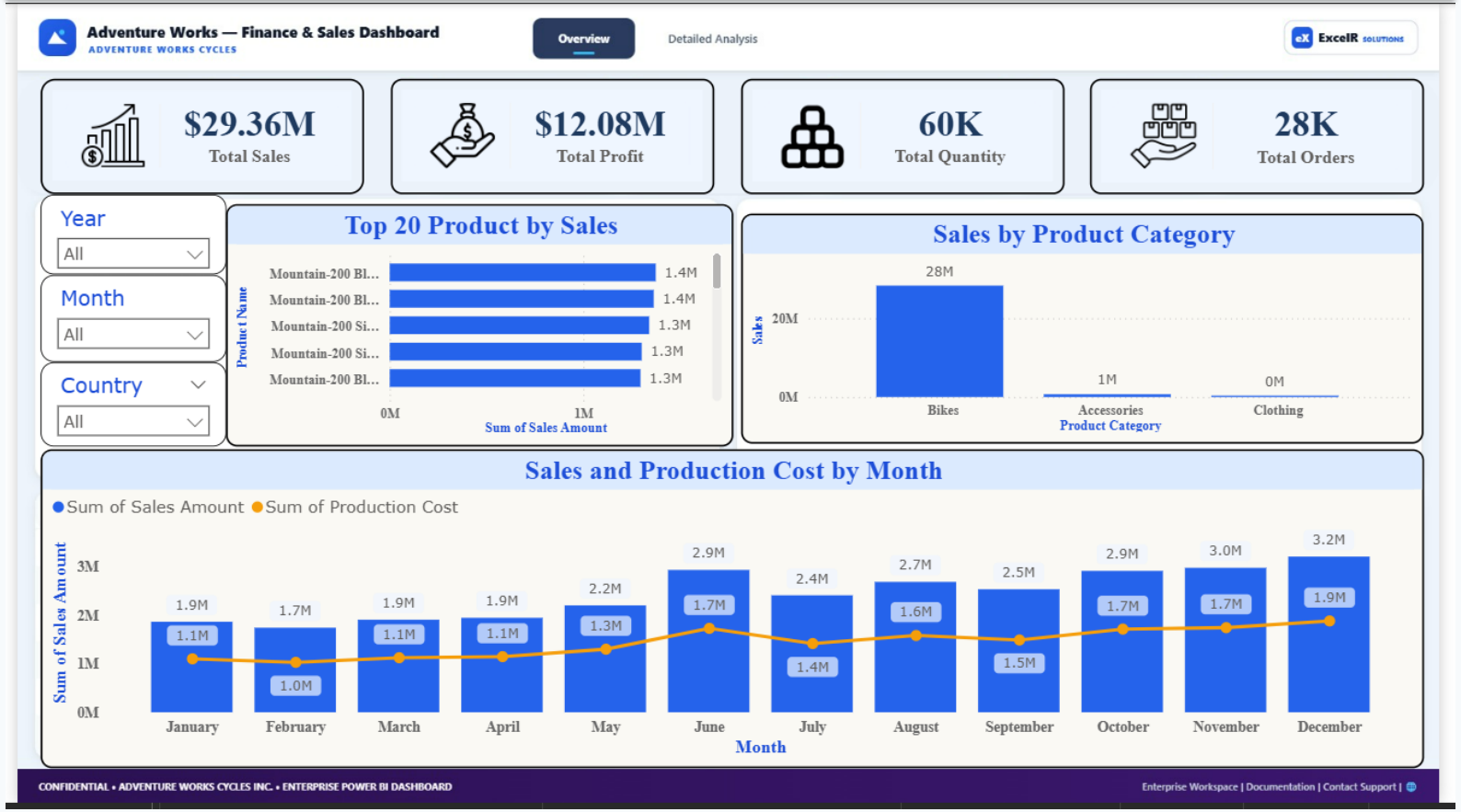
Charts

Bar, line, pie and combination charts for performance comparison.

Excel Work Completed
Data cleaning • calculated fields • pivot analysis • dashboard formatting

POWER BI DASHBOARD

PAGE 1 - FINANCE & SALES OVERVIEW



Executive KPIs

Monitors $29.36M Total Sales, $12.08M Total Profit, 60K Units, and 28K Total Orders.

Top Products & Categories

Bikes drive $28M revenue; Mountain-200 series leads top 20 selling products.

Sales vs Cost Trajectory

Monthly combo tracking revenue vs cost, peaking in Dec at $3.2M sales ($1.9M cost).

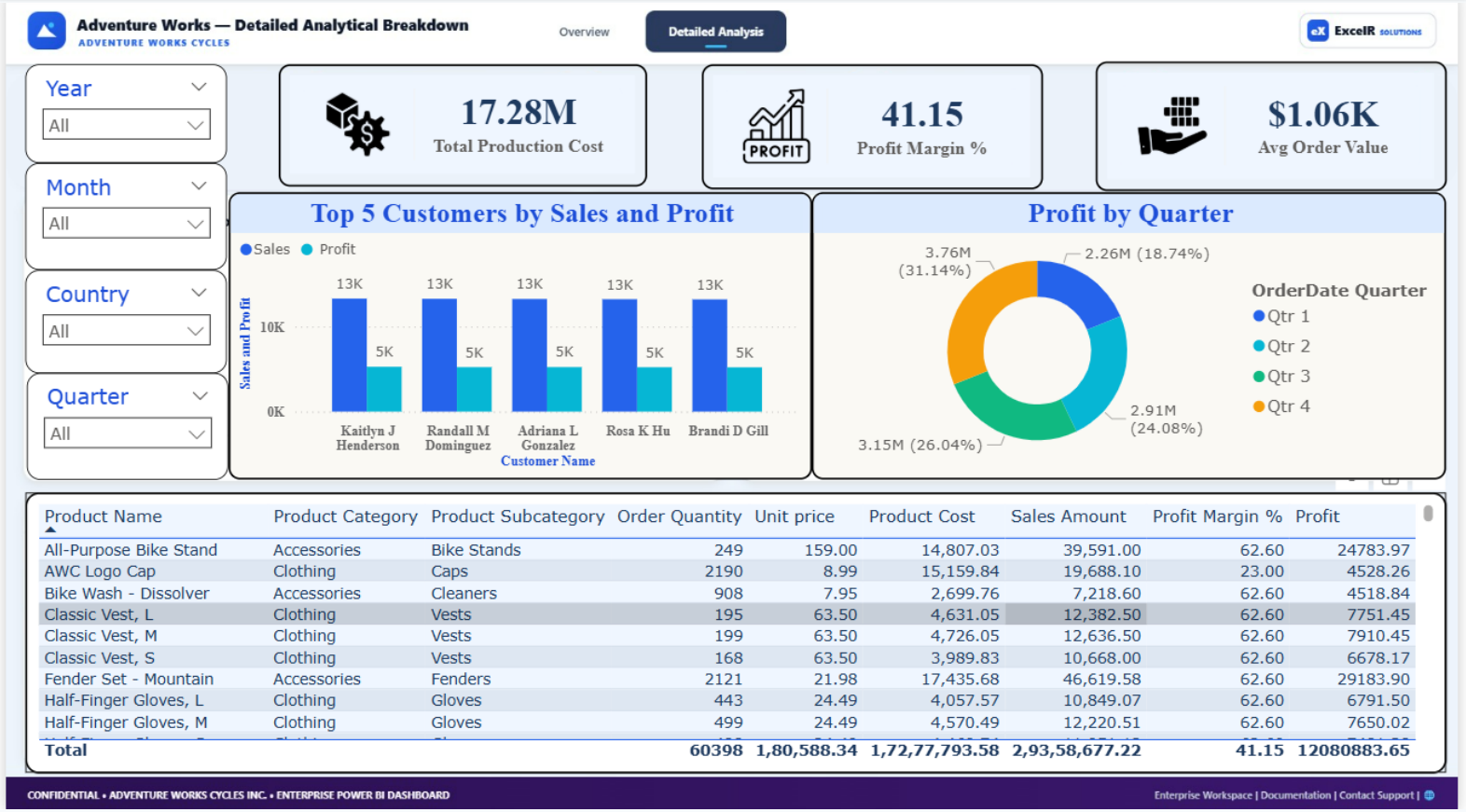
Interactive Slicers

Dynamic multi-level filtering by Year, Month, and Country across all overview visuals.

Power BI Overview Work Completed
Data Modeling • DAX Calculations • Monthly Trends & Slicers

POWER BI DASHBOARD

PAGE 2 - DETAILED ANALYTICAL BREAKDOWN



Product Name	Product Category	Product Subcategory	Order Quantity	Unit price	Product Cost	Sales Amount	Profit Margin %	Profit
All-Purpose Bike Stand	Accessories	Bike Stands	249	159.00	14,807.03	39,591.00	62.60	24783.97
AWC Logo Cap	Clothing	Caps	2190	8.99	15,159.84	19,688.10	23.00	4528.26
Bike Wash - Dissolver	Accessories	Cleaners	908	7.95	2,699.76	7,218.60	62.60	4518.84
Classic Vest, L	Clothing	Vests	195	63.50	4,631.05	12,382.50	62.60	7751.45
Classic Vest, M	Clothing	Vests	199	63.50	4,726.05	12,636.50	62.60	7910.45
Classic Vest, S	Clothing	Vests	168	63.50	3,989.83	10,668.00	62.60	6678.17
Fender Set - Mountain	Accessories	Fenders	2121	21.98	17,435.68	46,619.58	62.60	29183.90
Half-Finger Gloves, L	Clothing	Gloves	443	24.49	4,057.57	10,849.07	62.60	6791.50
Half-Finger Gloves, M	Clothing	Gloves	499	24.49	4,570.49	12,220.51	62.60	7650.02
Total			**60398**	**1,80,588.34**	**1,72,77,793.58**	**2,93,58,677.22**	**41.15**	**12080883.65**

Cost & Margin Metrics

Tracks $17.28M Production Cost, 41.15% Profit Margin, and $1.06K Avg Order Value.

Top 5 Customers Analysis

Identifies high-value customers (Kaitlyn Henderson, Randall Dominguez) by sales & profit.

Quarterly Profit Distribution

Quarterly donut breakdown showing Q4 leading with $3.76M (31.14%) profit contribution.

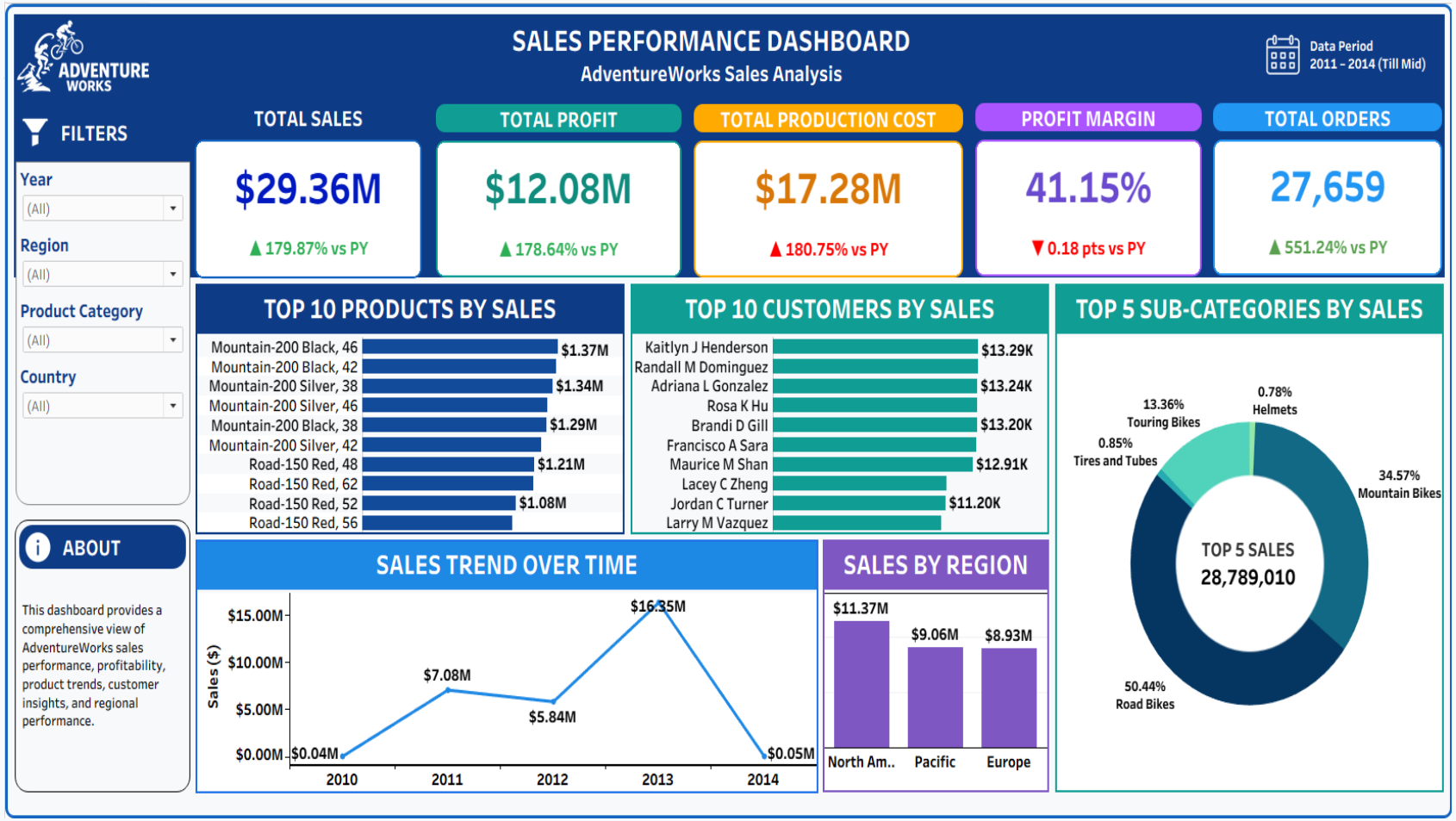
Granular Product Matrix

Detailed matrix table breaking down product details, order quantity, cost, margin, and profit.

Power BI Detailed Analysis Completed
Customer Level Analytics • Profit Share by Quarter • Matrix Table

TABLEAU DASHBOARD

TABLEAU DASHBOARD SCREENSHOT



DATA CONNECTION & PREPARATION

Connected and prepared AdventureWorks datasets using Tableau for analysis and visualization.

DATA MODELING

Created relationships between sales, customer, product, date and sales territory data for analysis.

CALCULATED FIELDS & KPIs

Created calculated fields for Sales, Profit, Production Cost, Profit Margin and other key metrics.

Interactive Dashboard

Interactive filters and visualizations for Year, Region, Product Category, Customer and Sales analysis.

Tableau Work Completed
Data Preparation • Calculated Fields • KPI Analysis • Interactive Visual Storytelling

SQL QUERIES - Q0 TO Q6

MySQL Transformations & KPIs

Q0 - Union of Both Sales Tables

```
CREATE TABLE Fact_Sales AS
SELECT * FROM FactInternetSales
UNION ALL SELECT * FROM Fact_Internet_Sales_New;
```

Q1 - Lookup Product Name

```
SELECT f.*, p.EnglishProductName FROM Fact_Sales f
LEFT JOIN DimProduct p ON f.ProductKey = p.ProductKey;
```

Q2 - Lookup Customer Name & Unit Price

```
SELECT f.*, CONCAT(c.FirstName, ' ', c.LastName) AS CustomerFullName, p.ListPrice
FROM Fact_Sales f LEFT JOIN DimProduct p ON f.ProductKey = p.ProductKey
LEFT JOIN DimCustomer c ON f.CustomerKey = c.CustomerKey;
```

Q3 - Date Columns from OrderDateKey

```
YEAR(...) AS Year, MONTH(...) AS MonthNo, MONTHNAME(...) AS MonthName,
CONCAT('Q', QUARTER(...)) AS Quarter, DATE_FORMAT(..., '%Y-%b') AS YearMonth,
CASE WHEN MONTH>=4 THEN MONTH-3 ELSE MONTH+9 END AS FinancialMonth
```

Q4 - Sales Amount

```
SELECT (UnitPrice * OrderQuantity * (1 - UnitPriceDiscountPct)) AS SalesAmount
FROM Fact_Sales;
```

Q5 - Production Cost & Q6 - Profit

```
SELECT (ProductStandardCost * OrderQuantity) AS ProductionCost FROM Fact_Sales;
SELECT (SalesAmount - ProductionCost) AS Profit FROM Fact_Sales;
```

SQL QUERIES - Q7 TO Q13

MySQL Transformations & KPIs

Q7 - Monthly Sales (Pivot)

```
SELECT YEAR(...) AS Year, MONTHNAME(...) AS Month,
ROUND(SUM(UnitPrice*Qty*(1-Disc))/1000,2) || ' K' AS TotalSales
FROM Fact_Sales GROUP BY Year, Month ORDER BY Year, Month;
```

Q8 - Year-wise Sales (Bar Chart)

```
SELECT YEAR(STR_TO_DATE(OrderDateKey, '%Y%m%d')) AS Year,
ROUND(SUM(UnitPrice*OrderQuantity*(1-UnitPriceDiscountPct))/1000,2) || ' K' AS
TotalSales
FROM Fact_Sales GROUP BY Year ORDER BY Year;
```

Q9 - Month-wise Sales (Line Chart)

```
SELECT Year, MonthNo, MonthName, ROUND(SUM(SalesAmount)/1000,2) || ' K' AS
TotalSales
FROM Fact_Sales GROUP BY Year, MonthNo, MonthName ORDER BY Year, MonthNo;
```

Q10 - Quarter-wise Sales (Pie Chart)

```
SELECT CONCAT('Q', QUARTER(STR_TO_DATE(OrderDateKey, '%Y%m%d'))) AS Quarter,
ROUND(SUM(UnitPrice*OrderQty*(1-Disc))/1000,2) || ' K' AS TotalSales
FROM Fact_Sales GROUP BY Quarter ORDER BY Quarter;
```

Q11 - Sales vs Production Cost (Combo)

```
SELECT Year, ROUND(SUM(SalesAmount)/1000,2) || ' K' AS Sales,
ROUND(SUM(ProductionCost)/1000,2) || ' K' AS ProductionCost
FROM Fact_Sales GROUP BY Year ORDER BY Year;
```

Q12 & Q13 - Overall Dashboard KPIs

```
SELECT COUNT(DISTINCT CustomerKey) AS TotalCustomers,
CONCAT(ROUND(SUM(SalesAmount)/1000000,2),' M') AS Total_Sales,
CONCAT(ROUND(SUM(Profit)/1000000,2),' M') AS Total_Profit FROM Fact_Sales;
```

POWER QUERY - M FORMULAS

Excel Power Query Data Prep Steps

Q0 - Union (Append Queries)

```
= Table.Combine({FactInternetSales, Fact_Internet_Sales_New})
```

Q1 - Product Name Lookup

```
= Table.NestedJoin(Fact_Sales, "ProductKey", DimProduct, "ProductKey", "DimProduct",
JoinKind.LeftOuter)
= Table.ExpandTableColumn(..., {"EnglishProductName"})
```

Q2 - Customer Name & Price Lookup

```
= Table.AddColumn(tbl, "CustomerFullName", each [FirstName] & " " &
([MiddleName]??"") & " " & [LastName])
= Table.NestedJoin(..., DimProduct, "ListPrice")
```

Q3 - Date Fields from OrderDateKey

```
= Table.AddColumn(tbl, "OrderDate", each Date.From(Text.From([OrderDateKey]),
"yyyyMMdd"))
= Table.AddColumn(..., "Year", each Date.Year([OrderDate]))
= Table.AddColumn(..., "FinancialMonth", each if Date.Month([OrderDate])>=4 then
Date.Month([OrderDate])-3 else Date.Month([OrderDate])+9)
```

Q4 & Q5 - SalesAmount & Cost

```
= Table.AddColumn(tbl, "SalesAmount", each [UnitPrice] * [OrderQuantity] * (1 -
([UnitPriceDiscountPct]??0)))
= Table.AddColumn(tbl, "ProductionCost", each [ProductStandardCost] *
[OrderQuantity])
```

Q6 - Profit Calculation

```
= Table.AddColumn(tbl, "Profit", each [SalesAmount] - [ProductionCost])
```

KEY INSIGHTS

29.36M	12.08M	41.15%	1,061.45
Total Sales	Total Profit	Profit Margin	Avg Order Value

1 Peak Year: 2013

2013 delivered the highest annual sales, clearly ahead of the surrounding years.

2 Top Region: Australia

Australia is the largest single region ($9.06M); Southwest leads the US regions ($5.72M).

3 Top Product

Mountain bike products dominate the top-selling list and drive a strong share of sales.

4 Top Customer

Top customers contribute significant individual order values and are useful targets for retention.

5 Quarter Trend

Quarter-level analysis highlights the strongest period and supports seasonal planning.

6 Sales > Cost

Production cost remains below sales, supporting a positive and healthy profit margin.

THANK YOU

Adventure Works Sales Analysis
Excel + Power BI

EXCEL

POWER BI

TABLEAU

SQL